

Francisco Sanchez

Senior Software Engineer

CONTACT

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Phone

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Location

Lasarte-Oria, Spain

CORE SKILLS

Team Leadership

System Architecture

Cybersecurity

.NET / ASP.NET

C++ / C#

JavaScript / TypeScript

Real-Time Systems

TECHNOLOGIES

C# · C++

.NET MAUI · WPF · WCF

ASP.NET Core · Entity FW

Docker · Git

RabbitMQ · gRPC · TFS

ROS / ROS2 · Linux

JavaScript · TypeScript

LANGUAGES

Spanish

Native

English

Proficient

EDUCATION

PhD — Information & Communications Technologies

Univ. Politécnica de Cartagena

Outstanding, Cum Laude · 2010–2017

MSc — Information & Communications Technologies

Univ. Politécnica de Cartagena

2009 – 2010

BEng — Electronics & Communications

INTEC, Dominican Republic

2005 – 2009

PROFESSIONAL PROFILE

Passionate and curious software engineer with over a decade of experience building robust, high-performance solutions across cybersecurity, backend systems, and real-time embedded software. Holding a PhD in Information & Communications Technologies (Outstanding, Cum Laude), I combine deep academic expertise with extensive industry practice — bringing rigorous design thinking and pragmatic engineering to every project. I care deeply about software quality, clean architecture, and continuous learning.

WORK EXPERIENCE

Engineering Manager · Senior Software Engineer

Aug 2021 – Present

SealPath Technologies · Bilbao, Spain

- ▶ Led and participated in the ongoing development and improvement of data-focused cybersecurity modules — covering encryption, digital rights management (DRM), information protection, and access traceability for enterprise clients.
- ▶ Coordinated cross-functional team efforts aimed at minimizing information leaks, guiding technical direction while remaining hands-on across new features and bug fixes.
- ▶ Leading product architecture definition across core company products, working closely with product team leads to align technical direction and design decisions.
- ▶ Designing and implementing an AI-driven development process across company squads, streamlining workflows and improving delivery through the integration of AI tooling into the engineering lifecycle.
- ▶ Preparing and delivering technical proposals for new products and features to both technical and non-technical stakeholders across the company.

Stack: C# · C++ · .NET MAUI · WPF · ASP.NET Core · gRPC · RabbitMQ · Docker · Hashicorp Nomad

Tech Lead · Senior Software Engineer

Dec 2017 – Jul 2021

Alerion Technologies · San Sebastián, Spain

- ▶ Served as technical lead for the redesign and evolution of the drone's control architecture, encompassing multi-sensor data capture (camera, LiDAR, altimeter, accelerometer), autonomous flight decision-making, and local processing and storage of enriched imagery (fused with GPS, LiDAR, and other data).
- ▶ Led model selection and fine-tuning of neural network-based machine learning models for object classification applied to blade damage detection using PyTorch. Originally a client-funded project, it led to the creation of the data analytics team; participate in the team's personnel selection process.
- ▶ Developed a GPU-accelerated image capture application from scratch (using an external library for gpu image processing), owning the full pipeline design and implementation, then transitioned to leading the team responsible for its continued evolution.
- ▶ Designed and implemented a state-machine library for use within the drone's control system, based on concepts developed during doctoral research.

Stack: C++ · ROS / ROS2 · Linux · Docker · Klepsydra · Git

Senior Software Engineer

Jun 2017 – Dec 2017

Salto Systems · Oiartzun, Spain

- ▶ Participated in the development and maintenance of microservices and applications for an intelligent cloud-based access control management platform.
- ▶ Collaborated on cloud infrastructure tasks, contributing to security and deployment aspects of the system.

Stack: Docker · Kubernetes · Jenkins · gRPC · Git

Software Engineer

Feb 2015 – Jun 2017

SealPath Technologies · Bilbao, Spain

- ▶ Contributed to the development and maintenance of core cybersecurity products — SealPath Desktop, File Server, and Security Sandbox — implementing new features and resolving issues across varying levels of complexity.

Stack: C# · C++ · WPF · WCF · Windows Forms · TFS

Software Engineer

Oct 2014 – Jan 2015

Cas-Training • Indra Software Labs · Lleida, Spain

- ▶ Took part in the migration of an inter-agency data interoperability platform from Weblogic Integration to Oracle SOA Suite (BPEL + OSB), contributing to service development and message transformation tasks.

Stack: Oracle SOA Suite · BPEL · OSB · XQuery

Research Fellow

Oct 2010 – Sep 2014

Universidad Politécnica de Cartagena · Cartagena, Spain

- ▶ Conducted research on hard real-time systems, model-driven engineering (MDE), and component-based software engineering (CBSE).
- ▶ Independently designed and developed a C/C++ framework for component-based real-time applications; co-authored 12+ conference and journal papers and a Springer book chapter.

Stack: C · C++ · MDE · CBSE · Real-Time Systems

PERSONAL PROJECTS

Rotawise — Fair & Balanced Doctor Scheduling

- ▶ Independently designed and built a constraint-based scheduling engine for fair distribution of doctor on-call rotas — hard rules (rest intervals, post-call, unit coverage) combined with a soft fairness score and weighted random sampling, solving a real need for a healthcare professional.
- ▶ In active, real-world use since 2025; PWA-enabled for offline use. Development directed through AI-assisted workflows, from requirements and domain logic through to deployment and maintenance.
- ▶ Architected as a fully client-side SPA with no backend — scheduling algorithm runs in a Web Worker, with local-first persistence via IndexedDB and the File System Access API.

Live: rotawise.fasli.dev

GitHub: github.com/fasaled/rota-wise

Stack: React · TypeScript · Vite · Tailwind CSS · IndexedDB · Web Workers · PWA · Playwright

Sailkari — Local LLM Classification Harness (Cross-Platform TUI + MCP)

- ▶ CLI/TUI that classifies text files with a local LLM, storing results portably (.sailkari/results.json) — offline, non-destructive, cross-platform (macOS, Linux, Windows).
- ▶ Designed an evaluation harness with swappable models and system prompts, tested systematically across five local models on a 24-file benchmark set including deliberately ambiguous cases.
- ▶ In-process inference engine (no subprocess/HTTP calls); full-screen terminal interface, or an MCP server over stdio from the same executable for agent-based clients; test suite with CI. Development directed through AI-assisted workflows, from requirements and domain logic through to deployment and maintenance.

npm: [@fasaled/sailkari](https://www.npmjs.com/~fasaled/sailkari)

GitHub: github.com/fasaled/sailkari

Stack: TypeScript, Node.js, llama.cpp

Muin — PDF Internal Structure Explorer (CLI + MCP + VS Code)

- ▶ Independently designed and built a tool that models a PDF's internal structure — indirect objects, streams, page tree, cross-references — as a navigable graph, with one command core powering an interactive TUI, an MCP server for AI agents, and a VS Code extension with no duplicated logic.
- ▶ Built on qpdf compiled to WebAssembly for low-level PDF parsing; explicit limits on file size, object count, and graph export depth.
- ▶ Published across three distribution channels — npm, the VS Code Marketplace, and GitHub. Development directed through AI-assisted workflows, with architectural decisions recorded as they were made.

GitHub: github.com/fasaled/muin

Stack: TypeScript, Ink, WebAssembly, MCP

SELECTED PUBLICATIONS

Journals: Robotica (Cambridge, 2014) · The Scientific World Journal (2014) · RIAI (2015)

Book Chapter: Embedded and Real Time System Development — Springer (2014)

Conferences: 12+ papers at ICSoft, SISTEDES, Jornadas de Tiempo Real, and others.